

Chapter 6: Mechanical Properties of Matter

Key Concepts

Elasticity — Property of a body to return to its original shape/size after deforming force is removed.

Plasticity — Property of a body to retain the deformed shape permanently after deforming force is removed.

Stress — Deforming force per unit area. $\sigma = F/A$ (N m⁻² or Pa).

Strain — Ratio of change in dimension to original dimension (dimensionless). Three types: tensile, compressive, shear.

Young's Modulus (Y) — Ratio of tensile/compressive stress to longitudinal strain. $Y = \frac{F/A}{\Delta L/L} = \frac{FL}{A\Delta L}$

Hooke's Law — Within elastic limit, extension in a spring is directly proportional to applied force. $F = kx$ (k = spring constant).

Spring Constant (k) — Force per unit extension. $k = F/x$ (unit: Nm^{-1}).

Pressure — Force acting normally per unit area. $P = F/A$ (unit: $\text{Nm}^{-2} = \text{Pa}$).

Atmospheric Pressure — Pressure exerted by the Earth's atmosphere. 1 atm = 1.013×10^5 Pa.

Density — Mass per unit volume. $\rho = \frac{m}{V}$ — C kgm³).

Archimedes Principle — Upward buoyant force equals weight of fluid displaced.

Principle of Floatation — A floating body displaces its own weight of fluid.

Pascal's Law — Pressure applied to an enclosed fluid is transmitted equally in all directions.

Hydraulic Press — Device based on Pascal's law. Magnifies force: $F_2/F_1 = A_2/A_1$.

Viscosity — Internal friction between fluid layers opposing relative motion.

Terminal Velocity — Constant velocity reached when drag force equals weight of falling object.

Stokes Law — Drag force on a sphere moving through a viscous fluid: $F = 6\pi \cdot r \cdot v$.

[illegible]

MCQ's

1. The property of a body to regain its original shape after removal of deforming force is called:

A. Plasticity B. Elasticity C. Ductility D. Malleability

Answer: B

2. SI unit of stress is:

A. Nm B. Nm {¹ C. Nm {² D. Nm²

Answer: C

3. Hooke's law holds up to:

A. Breaking point B. Elastic limit C. Yield point D. Plastic limit

Answer: B

4. Young's modulus is defined for which type of strain?

A. Shear strain B. Volumetric strain C. Longitudinal strain D. None

Answer: C

5. The spring constant of a spring is 200 Nm^{-1} . The force needed to extend it by 5 cm is:

A. 10 N B. 20 N C. 40 N D. 1000 N

Answer: A ($F = kx = 200 \times 0.05 = 10 \text{ N}$)

6. Atmospheric pressure at sea level is approximately:

A. 1.013×10^4 Pa B. 1.013×10^5 Pa C. 1.013×10^6 Pa D. 1.013×10^3 Pa

Answer: B

7. Which law states that pressure applied to an enclosed fluid is transmitted equally in all directions?

A. Hooke's law B. Archimedes principle C. Pascal's law D. Stokes law

Answer: C

8. The upward force exerted by a fluid on a submerged object is called:

A. Thrust B. Pressure C. Buoyant force D. Viscous force

Answer: C

9. A floating body displaces fluid equal to its own:

A. Volume B. Density C. Weight D. Mass

Q9. Define terminal velocity.

Ans: Terminal velocity is the constant maximum velocity attained by a falling object when the upward drag force (viscous resistance) balances the downward weight of the object, resulting in zero net acceleration.

Q10. What is the difference between density and relative density?

Ans: Density ($\rho = \frac{m}{V}$) is the mass per unit volume of a substance (ρ has SI unit kg m^{-3}). Relative density (also called specific gravity) is the ratio of density of a substance to density of water at 4°C . It is dimensionless and has no unit.

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Long Questions

Q1. Derive the expression for Young's modulus. Describe an experiment to determine it for a wire.

Ans: Young's modulus (Y) is defined as the ratio of longitudinal stress to longitudinal strain within the elastic limit.

Consider a wire of length L and cross-sectional area A . A force F is applied along its length, causing an extension ΔL .

Stress = F/A

Strain = $\Delta L/L$

Young's Modulus: $Y = \text{Stress} / \text{Strain} = (F/A) / (\Delta L/L)$

Experiment (Searle's apparatus):

1. Take two identical long wires (usually steel) suspended from a rigid support.
2. Attach a main scale and vernier scale arrangement between the two wires.
3. The reference wire carries a fixed load; the experimental wire carries a load that can be varied.
4. Add weights to the experimental wire gradually and note the extension using the vernier scale.
5. Measure the length L with a meter scale, diameter d with a screw gauge ($A = \pi d^2/4$).
6. Plot a stress-strain graph — it is a straight line up to the elastic limit.
7. Calculate Y from the slope of the graph using $Y = FL / A\Delta L$.

Q2. State and explain Hooke's law. Draw the stress-strain graph and explain its different regions.

Ans: Hooke's law: Within the elastic limit, stress is directly proportional to strain for a given material. $\sigma \propto \epsilon$ or $\sigma = Y\epsilon$

Stress-Strain Graph:

- **OA (Proportional limit):** Stress $\propto$ Strain, Hooke's law holds, graph is linear.
- **AB (Elastic region):** Graph deviates from linearity but body still returns to original shape when load is removed. Point B is the elastic limit.
- **BC (Plastic region):** Beyond elastic limit, permanent deformation occurs. The material yields without much increase in stress.
- **CD (Strain hardening):** Further stress increases strain at a slower rate due to rearrangement of internal structure.
- **Point D (Ultimate tensile strength):** Maximum stress the material can withstand.
- **DE (Necking):** Cross-section decreases locally, stress decreases, until fracture at point E (breaking point).

Q3. Explain Pascal's law. Describe the working of a hydraulic press with a diagram.

Ans: Pascal's law: Pressure applied to an enclosed fluid is transmitted equally and undiminished to every part of the fluid and the walls of the container.

Hydraulic Press Construction:

A hydraulic press consists of two cylinders of different cross-sectional areas connected by a pipe filled with an incompressible fluid (usually oil). Each cylinder has a piston — a small piston of area A_1 and a large piston of area A_2 .

Working:

1. A small force F_1 is applied on the small piston.
2. According to Pascal's law, the pressure $P = F_1/A_1$ is transmitted equally throughout the fluid.
3. The same pressure P acts on the large piston: $P = F_2/A_2$.
4. Therefore, $F_2 = P \times A_2 = (F_1/A_1) \times A_2$.
5. Since $A_2 > A_1$, the output force F_2 is magnified by the factor A_2/A_1 .

Thus a small input force produces a large output force, making the hydraulic press a force-multiplying machine.

Q4. Derive Stokes law and explain how terminal velocity is determined for a sphere falling through a

viscous fluid.

Ans: Stokes Law:

When a sphere of radius r moves through a viscous fluid of viscosity η , velocity v , the drag force F is given by:
 $F = 6\pi\eta rv$

Terminal Velocity Derivation:

Consider a sphere of radius r and density ρ_s falling through a fluid of density ρ_f

Three forces act on the sphere:

Weight (downward): $W = \frac{4}{3}\pi r^3 \rho_s g$

Buoyant force (upward): $U = \frac{4}{3}\pi r^3 \rho_f g$

Viscous drag (upward): $F = 6\pi\eta rv$

Initially, the sphere accelerates. As velocity increases, drag force increases. When net force becomes zero, the sphere falls with constant velocity (terminal velocity v_t).

At terminal velocity:

$$W - U - F = 0$$

$$\frac{4}{3}\pi r^3 \rho_s g - \frac{4}{3}\pi r^3 \rho_f g - 6\pi\eta rv_t = 0$$

$$\frac{4}{3}\pi r^3 (\rho_s - \rho_f) g = 6\pi\eta r v_t$$

Solving for v_t :

$$v_t = \frac{2r^2 g (\rho_s - \rho_f)}{9\eta}$$

This shows terminal velocity $\propto r^2$, $\propto g$, $\propto (\rho_s - \rho_f)$, $\propto 1/\eta$

Q5. Explain the principle of floatation with examples. Why does a ship made of iron float while an iron nail sinks?

Ans: Principle of Floatation:

A floating body displaces its own weight of the fluid in which it floats. Mathematically, Weight of body = Weight of fluid displaced.

Conditions for Floatation:

1. If $\rho_{\text{body}} < \rho_{\text{fluid}}$: Body floats partially submerged.
2. If $\rho_{\text{body}} = \rho_{\text{fluid}}$: Body floats completely submerged at any depth.
3. If $\rho_{\text{body}} > \rho_{\text{fluid}}$: Body sinks.

Why a ship floats but an iron nail sinks:

A ship is made of iron (density $\sim 7800 \text{ kg m}^{-3}$, much denser than water $\sim 1000 \text{ kg m}^{-3}$). However, the ship is hollow and contains large air spaces. The average density of the ship (total mass / total volume including air spaces) becomes less than that of water. Therefore, it displaces enough water to equal its own weight and floats.

An iron nail is solid with no air spaces. Its density (7800 kg m^{-3}) is greater than water (1000 kg m^{-3}), so its weight exceeds the buoyant force, and it sinks.

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Numerical Problems

Q1. A spring extends by 3 cm when a force of 6 N is applied. Find its spring constant and the extension for a force of 15 N.

Solution:

Given: $x = 3 \text{ cm} = 0.03 \text{ m}$, $F = 6 \text{ N}$, $F_2 = 15 \text{ N}$

Spring constant: $k = F/x = 6/0.03 = 200 \text{ Nm}^{-1}$

For $F_2 = 15 \text{ N}$: $x_2 = F_2/k = 15/200 = 0.075 \text{ m} = 7.5 \text{ cm}$

Q2. A steel wire of length 2 m and cross-sectional area $1 \times 10^{-6} \text{ m}^2$ stretches by 0.5 mm under a load of 100 N. Calculate Young's modulus of steel.

Solution:

Given: $L = 2 \text{ m}$, $A = 1 \times 10^{-6} \text{ m}^2$, $\Delta L = 0.5 \text{ mm} = 0.5 \times 10^{-3} \text{ m}$

Stress = $F/A = 100 / (1 \times 10^{-6}) = 1 \times 10^8 \text{ Pa}$

Strain = $\Delta L/L = 0.5 \times 10^{-3} / 2 = 2.5 \times 10^{-4}$

$Y = \text{Stress}/\text{Strain} = 1 \times 10^8 / 2.5 \times 10^{-4} = 4 \times 10^{11} \text{ Pa}$

Q3. A force of 50 N acts normally on a surface of area 0.02 m². Calculate the pressure exerted.

Solution:

Given: $F = 50 \text{ N}$, $A = 0.02 \text{ m}^2$

$$P = F/A = 50/0.02 = 2500 \text{ Pa} = 2.5 \times 10^3 \text{ Pa}$$

Q4. The small piston of a hydraulic press has an area of 0.005 m² and the large piston has an area of 0.1 m². If a force of 100 N is applied on the small piston, calculate the force on the large piston.

Solution:

Given: $A_1 = 0.005 \text{ m}^2$, $A_2 = 0.1 \text{ m}^2$, $F_1 = 100 \text{ N}$

$$\text{Pressure } P = F_1/A_1 = 100/0.005 = 20000 \text{ Pa}$$

$$\text{Force on large piston: } F_2 = P \times A_2 = 20000 \times 0.1 = 2000 \text{ N}$$

$$\text{Alternatively, } F_2 = F_1 \times (A_2/A_1) = 100 \times (0.1/0.005) = 100 \times 20 = 2000 \text{ N}$$

Q5. The density of gold is 19300 kgm⁻³ and volume of a gold bar is 0.0005 m³. Calculate its mass.

Solution:

Given: $\rho = 19300 \text{ kgm}^{-3}$, $V = 0.0005 \text{ m}^3$

$$m = \rho V = 19300 \times 0.0005 = 9.65 \text{ kg}$$

Q6. A cube of side 0.1 m is completely immersed in water. Calculate the buoyant force acting on it. (Density of water = 1000 kgm⁻³, $g = 10 \text{ ms}^{-2}$)

Solution:

Given: side = 0.1 m, $\rho_{\text{water}} = 1000 \text{ kgm}^{-3}$, $g = 10 \text{ ms}^{-2}$

$$\text{Volume of cube} = (0.1)^3 = 0.001 \text{ m}^3$$

$$\text{Weight of water displaced} = \rho V g = 1000 \times 0.001 \times 10 = 10 \text{ N}$$

$$\text{Buoyant force} = \text{Weight of water displaced} = 10 \text{ N}$$

Q7. A steel ball of radius 2 mm falls through glycerine. Density of steel = 7800 kgm⁻³, density of glycerine = 1260 kgm⁻³, viscosity of glycerine = 0.83 Pas. Calculate the terminal velocity. ($g = 10 \text{ ms}^{-2}$)

Solution:

Given: $r = 2 \times 10^{-3} \text{ m}$, $\rho_{\text{steel}} = 7800 \text{ kgm}^{-3}$, $\rho_{\text{glycerine}} = 1260 \text{ kgm}^{-3}$, $\eta = 0.83 \text{ Pa s}$, $g = 10 \text{ ms}^{-2}$

$$v_t = \frac{2r^2g(\rho_{\text{steel}} - \rho_{\text{glycerine}})}{9\eta}$$

$$v_t = \frac{2 \times (2 \times 10^{-3})^2 \times 10 \times (7800 - 1260)}{9 \times 0.83}$$

$$v_t = \frac{2 \times 4 \times 10^{-6} \times 10 \times 6540}{9 \times 0.83}$$

$$v_t = \frac{2 \times 4 \times 10^{-6} \times 10 \times 6540}{7.47}$$

$$v_t = 0.5232 / 7.47 = 0.07 \text{ ms}^{-1} \text{ (approximately)}$$

Q8. A spring of spring constant 500 Nm⁻¹ is compressed by 0.02 m. Calculate the force stored in the spring and the work done.

Solution:

Given: $k = 500 \text{ Nm}^{-1}$, $x = 0.02 \text{ m}$

$$\text{Force: } F = kx = 500 \times 0.02 = 10 \text{ N}$$

$$\text{Work done} = \text{Elastic potential energy} = \frac{1}{2}kx^2 = 0.5 \times 500 \times (0.02)^2$$

$$= 0.5 \times 500 \times 0.0004 = 0.5 \times 0.2 = 0.1 \text{ J}$$